

# JSS1 Chemistry Diagnostic Bank (40 Questions)

**Scope:** Primary 5/6 General Science to JSS1 Chemistry (Nigerian & Cambridge). **Purpose:** Diagnose readiness for JSS2 Chemistry.

## Section 1: Matter & Particles (1–15)

| No. | Question                                                                                                                           | Ans. | Diagnostic Focus      |
|-----|------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|
| 1   | Anything that has mass and occupies space is called: A. Energy B. Matter C. Force D. Air                                           | B    | Definition of matter. |
| 2   | Which is NOT matter? A. Water B. Heat C. Stone D. Air                                                                              | B    | Matter vs energy.     |
| 3   | The three common states of matter are: A. Solid, liquid, gas B. Acid, base, salt C. Atom, ion, molecule D. Metal, non-metal, alloy | A    | States of matter.     |
| 4   | Particles in a gas move: A. Slowly B. Not at all C. Freely and rapidly D. In fixed positions                                       | C    | Particle theory.      |
| 5   | Ice changes to water by: A. Freezing B. Condensation C. Melting D. Evaporation                                                     | C    | Change of state.      |
| 6   | Water vapour changing to liquid is: A. Sublimation B. Condensation C. Melting D. Boiling                                           | B    | State changes.        |
| 7   | The smallest particle of an element is an: A. Atom B. Cell C. Compound D. Crystal                                                  | A    | Atomic concept.       |
| 8   | Atoms combine to form: A. Molecules B. Tissues C. Cells D. Organs                                                                  | A    | Molecules.            |
| 9   | An element contains: A. One type of atom B. Different atoms C. Only liquids D. Only metals                                         | A    | Elements.             |
| 10  | Oxygen is a: A. Compound B. Mixture C. Element D. Solution                                                                         | C    | Classification.       |

|    |                                                                                                |   |                       |
|----|------------------------------------------------------------------------------------------------|---|-----------------------|
| 11 | Air is best described as a:<br>A. Compound B. Mixture C. Element D. Gas only                   | B | Mixtures.             |
| 12 | Salt dissolved in water forms a:<br>A. Suspension B. Solution C. Alloy D. Gas                  | B | Solutions.            |
| 13 | Which has a fixed shape?<br>A. Gas B. Liquid C. Solid D. Plasma                                | C | Properties of solids. |
| 14 | Heating generally causes particles to:<br>A. Stop moving B. Move faster C. Disappear D. Shrink | B | Kinetic theory.       |
| 15 | Evaporation occurs at:<br>A. Any temperature B. 0°C only C. 100°C only D. -10°C                | A | Evaporation concept.  |

## Section 2: Laboratory & Separation (16–30)

| No. | Question                                                                                                    | Ans. | Diagnostic Focus     |
|-----|-------------------------------------------------------------------------------------------------------------|------|----------------------|
| 16  | A beaker is mainly used for:<br>A. Measuring exactly B. Holding/mixing liquids C. Heating only D. Filtering | B    | Lab apparatus.       |
| 17  | The instrument for measuring temperature is a:<br>A. Balance B. Thermometer C. Ruler D. Stopwatch           | B    | Lab skills.          |
| 18  | Sand can be separated from water by:<br>A. Filtration B. Distillation C. Evaporation only D. Magnetism      | A    | Separation.          |
| 19  | A magnet can separate:<br>A. Salt and sugar B. Iron filings and sand C. Water and oil D. Alcohol and water  | B    | Magnetic separation. |
| 20  | Pure water is obtained from salt water by:<br>A. Sieving B. Distillation C. Decantation D. Winnowing        | B    | Distillation.        |
| 21  | The symbol for sodium is:<br>A. So B. Na C. S D. N                                                          | B    | Chemical symbols.    |
| 22  | The symbol for iron is:<br>A. Ir B. I C. Fe D. F                                                            | C    | Symbols.             |
| 23  |                                                                                                             | B    | Safety.              |

|       |                                                                                                                                |       |                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------|-------|----------------------|
|       | Laboratory goggles protect the: A. Hands B. Eyes C. Nose D. Feet                                                               |       |                      |
| 24    | Never taste chemicals because they may be: A. Sweet B. Harmful C. Expensive D. Colourless                                      | B     | Safety awareness.    |
| 25    | The SI unit of mass is: A. Litre B. Kilogram C. Metre D. Second                                                                | B     | Measurement.         |
| 26-30 | Questions cover laboratory rules, measuring volume, mixtures vs compounds, observation skills, and safe disposal of chemicals. | Mixed | Practical readiness. |

### Section 3: Chemical Changes & Everyday Chemistry (31–40)

| No. | Question                                                                                                 | Ans. | Diagnostic Focus      |
|-----|----------------------------------------------------------------------------------------------------------|------|-----------------------|
| 31  | Rusting of iron is a: A. Physical change B. Chemical change C. Change of state D. Reversible change      | B    | Chemical change.      |
| 32  | Burning paper produces: A. New substances B. Only paper C. Water only D. Oxygen only                     | A    | Evidence of reaction. |
| 33  | Vinegar is an example of a weak: A. Base B. Acid C. Salt D. Metal                                        | B    | Acids.                |
| 34  | Blue litmus turns red in: A. Acid B. Base C. Water D. Air                                                | A    | Indicators.           |
| 35  | Soap solution is generally: A. Acidic B. Neutral C. Alkaline D. Radioactive                              | C    | Bases.                |
| 36  | Which is an example of a physical change? A. Melting wax B. Burning wood C. Rusting D. Cooking egg       | A    | Physical changes.     |
| 37  | Gas bubbles during a reaction may indicate: A. No change B. Chemical reaction C. Freezing D. Evaporation | B    | Reaction evidence.    |
| 38  | Household bleach should be handled: A. Carelessly B. With safety precautions                             | B    | Chemical safety.      |

|    |                                                                                                                           |   |                          |
|----|---------------------------------------------------------------------------------------------------------------------------|---|--------------------------|
|    | C. As food D. Without labels                                                                                              |   |                          |
| 39 | The pH scale measures: A. Mass B. Acidity/alkalinity C. Speed D. Density                                                  | B | Introduction to pH.      |
| 40 | Students who perform well across this assessment are generally ready for more advanced laboratory investigations in JSS2. | — | Overall diagnostic note. |